

Class 14: RNASeq mini-project

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Background

In today's mini project The data for our hands-on session comes from GEO entry: GSE37704, which is associated with the following publication:

Trapnell C, Hendrickson DG, Sauvageau M, Goff L et al. "Differential analysis of gene regulation at transcript resolution with RNA-seq". Nat Biotechnol 2013 Jan;31(1):46-53. PMID: 23222703

The authors report on differential analysis of lung fibroblasts in response to loss of the developmental transcription factor HOXA1.

Data Import

Read counts and metadata CSV files

```
library(DESeq2)
```

Loading required package: S4Vectors

Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics

Attaching package: 'generics'

The following objects are masked from 'package:base':

```
as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,  
setequal, union
```

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

```
IQR, mad, sd, var, xtabs
```

The following objects are masked from 'package:base':

```
anyDuplicated, aperm, append, as.data.frame, basename, cbind,  
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,  
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,  
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,  
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,  
unsplit, which.max, which.min
```

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

```
metaFile <- "GSE37704_metadata.csv"
countFile <- "GSE37704_featurecounts.csv"

# Import metadata and take a peek
colData = read.csv(metaFile, row.names=1)
head(colData)
```

```
              condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

Sanity check

```
colData
```

```

condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369 hoxa1_kd
SRR493370 hoxa1_kd
SRR493371 hoxa1_kd

```

```

# Import countdata
countData = read.csv(countFile, row.names=1)
head(countData)

```

```

length SRR493366 SRR493367 SRR493368 SRR493369 SRR493370
ENSG00000186092 918 0 0 0 0 0
ENSG00000279928 718 0 0 0 0 0
ENSG00000279457 1982 23 28 29 29 28
ENSG00000278566 939 0 0 0 0 0
ENSG00000273547 939 0 0 0 0 0
ENSG00000187634 3214 124 123 205 207 212
SRR493371
ENSG00000186092 0
ENSG00000279928 0
ENSG00000279457 46
ENSG00000278566 0
ENSG00000273547 0
ENSG00000187634 258

```

Hmm... remember that we need the `countData` and `colData` files to match up so we will need to remove that odd first column in `countData` namely `countData$length`.

Q. Complete the code below to remove the troublesome first column from `countData`

```

countData <- as.matrix(countData[, -1])
head(countData)

```

```

SRR493366 SRR493367 SRR493368 SRR493369 SRR493370 SRR493371
ENSG00000186092 0 0 0 0 0 0
ENSG00000279928 0 0 0 0 0 0
ENSG00000279457 23 28 29 29 28 46
ENSG00000278566 0 0 0 0 0 0
ENSG00000273547 0 0 0 0 0 0
ENSG00000187634 124 123 205 207 212 258

```

Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0 read count across all samples (i.e. columns).

Tip: What will rowSums() of countData return and how could you use it in this context?

Setup DESeq object

```
dim(countData)
```

```
[1] 19808      6
```

```
dds = DESeqDataSetFromMatrix(countData= countData,  
                              colData= colData,  
                              design=~ condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds = DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
dds
```

```

class: DESeqDataSet
dim: 19808 6
metadata(1): version
assays(4): counts mu H cooks
rownames(19808): ENSG00000186092 ENSG00000279928 ... ENSG00000277475
               ENSG00000268674
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(2): condition sizeFactor

```

Run DESeq analysis pipeline

```
res = results(dds)
```

Extract the results

Big table with log2 fold changes and p-values

Q. Call the `summary()` function on your results to get a sense of how many genes are up or down-regulated at the default 0.1 p-value cutoff.

```
summary(res)
```

```

out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4393, 27%
outliers [1]      : 0, 0%
low counts [2]    : 1221, 7.6%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results

```

Data Vis

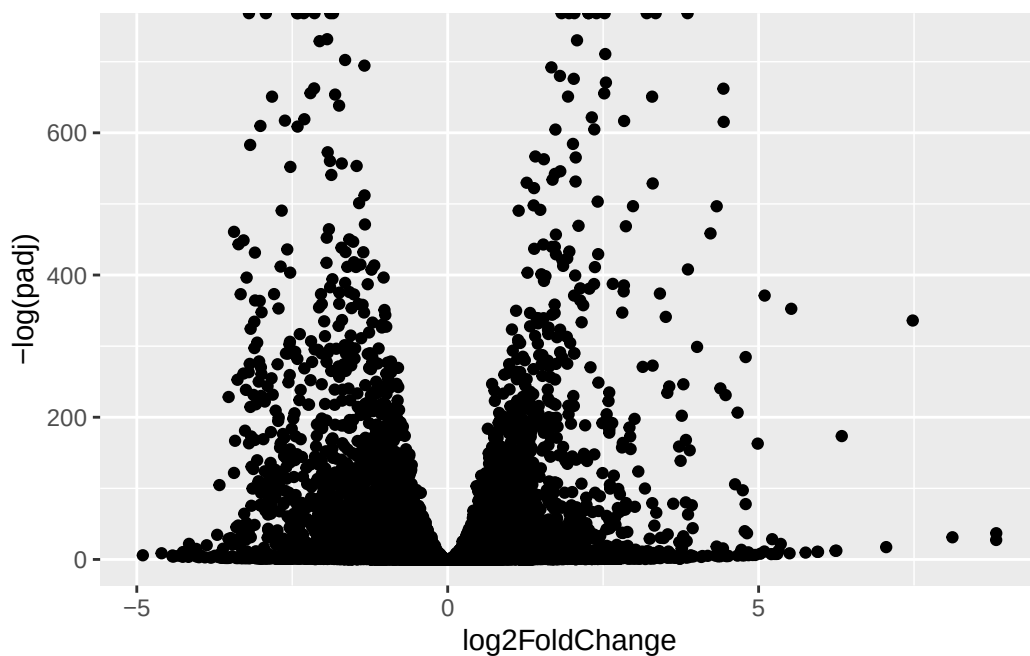
Volcano plot

Now we will make a volcano plot, a commonly produced visualization from this type of data that we introduced last day. Basically it's a plot of log2 fold change vs -log adjusted p-value.

```
library(ggplot2)

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point()
```

Warning: Removed 5054 rows containing missing values or values outside the scale range (`geom\_point()`).



Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff lines:


```

mycols <- rep("gray", nrow(res))

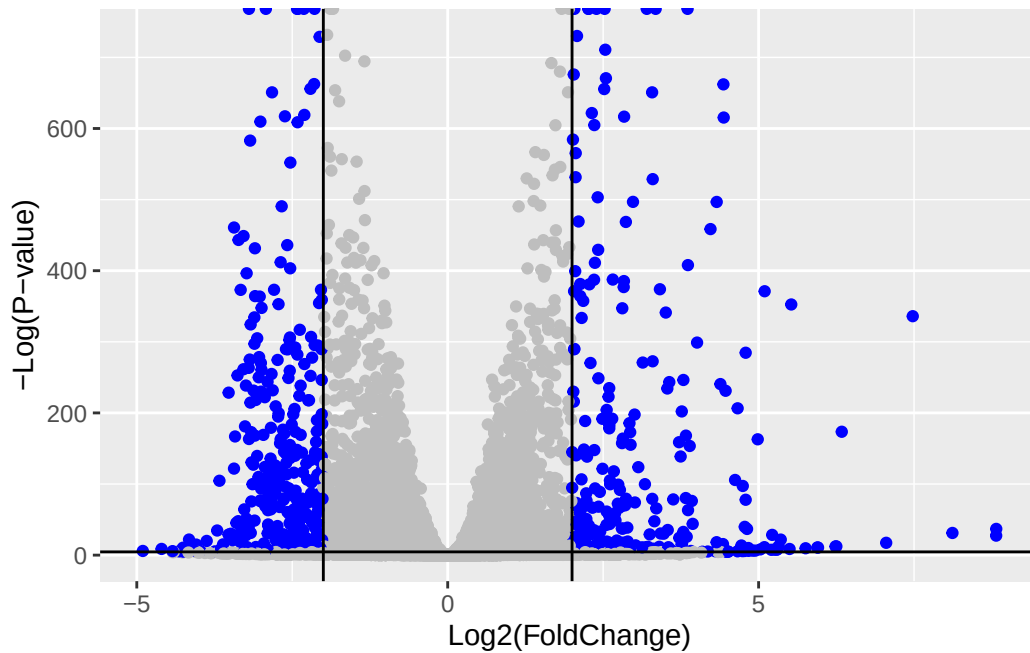
mycols[ abs(res$log2FoldChange) > 2 ] <- "blue"

mycols[ res$padj > 0.01 ] <- "gray"

ggplot(res) +
  aes(log2FoldChange,
      -log(padj)) +
  geom_point(col = mycols) +
  xlab("Log2(FoldChange)") +
  ylab("-Log(P-value)") +
  geom_vline(xintercept = c(-2, 2)) +
  geom_hline(yintercept = -log(0.01))

```

Warning: Removed 5054 rows containing missing values or values outside the scale range (`geom\_point()`).



Add annotation data

Since we mapped and counted against the Ensembl annotation, our results only have information about Ensembl gene IDs. However, our pathway analysis downstream will use KEGG pathways, and genes in KEGG pathways are annotated with Entrez gene IDs. So let's add them as we did the last day.

Q. Use the `mapIds()` function multiple times to add `SYMBOL`, `ENTREZID` and `GENENAME` annotation to our results by completing the code below.

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

```
columns(org.Hs.eg.db)
```

[1]	"ACCNUM"	"ALIAS"	"ENSEMBL"	"ENSEMBLPROT"	"ENSEMBLTRANS"
[6]	"ENTREZID"	"ENZYME"	"EVIDENCE"	"EVIDENCEALL"	"GENENAME"
[11]	"GENETYPE"	"GO"	"GOALL"	"IPI"	"MAP"
[16]	"OMIM"	"ONTOLOGY"	"ONTOLOGYALL"	"PATH"	"PFAM"
[21]	"PMID"	"PROSITE"	"REFSEQ"	"SYMBOL"	"UCSCKG"
[26]	"UNIPROT"				

```
res$symbol <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     keytype="ENSEMBL",
                     column= "SYMBOL",
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez <- mapIds(org.Hs.eg.db,
                     keys=row.names(res),
                     keytype="ENSEMBL",
                     column="ENTREZID",
                     multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
res$name <- mapIds(org.Hs.eg.db,
  keys=row.names(res),
  keytype= "ENSEMBL",
  column= "GENENAME",
  multiVals="first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res, 10)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000186092	0.0000	NA	NA	NA	NA
ENSG00000279928	0.0000	NA	NA	NA	NA
ENSG00000279457	29.9136	0.1792571	0.3248215	0.551863	5.81042e-01
ENSG00000278566	0.0000	NA	NA	NA	NA
ENSG00000273547	0.0000	NA	NA	NA	NA
ENSG00000187634	183.2296	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.1881	-0.6927205	0.0548465	-12.630156	1.43993e-36
ENSG00000187961	209.6379	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.2551	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.9798	0.5428105	0.5215598	1.040744	2.97994e-01

	padj	symbol	entrez	name
	<numeric>	<character>	<character>	<character>
ENSG00000186092	NA	OR4F5	79501	olfactory receptor f..
ENSG00000279928	NA	NA	NA	NA
ENSG00000279457	6.87080e-01	NA	NA	NA
ENSG00000278566	NA	NA	NA	NA
ENSG00000273547	NA	NA	NA	NA
ENSG00000187634	5.16278e-03	SAMD11	148398	sterile alpha motif ..
ENSG00000188976	1.76744e-35	NOC2L	26155	NOC2 like nucleolar ..
ENSG00000187961	1.13536e-07	KLHL17	339451	kelch like family me..
ENSG00000187583	9.18988e-01	PLEKHN1	84069	pleckstrin homology ..
ENSG00000187642	4.03817e-01	PERM1	84808	PPARGC1 and ESRR ind..

Q. Finally for this section let's reorder these results by adjusted p-value and save them to a CSV file in your current project directory.

```
res = res[order(res$padj),]
write.csv(res, file="deseq_results.csv")
```

Pathway analysis

KEGG, GO, and REACTOME

Here we are going to use the gage package for pathway analysis. Once we have a list of enriched pathways, we're going to use the pathview package to draw pathway diagrams, shading the molecules in the pathway by their degree of up/down-regulation.

```
library(pathview)
```

```
#####
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.
```

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

```
#####
```

```
library(gage)
```

```
library(gageData)

data(kegg.sets.hs)
data(sigmet.idx.hs)

# Focus on signaling and metabolic pathways only
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]

# Examine the first 3 pathways
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

```
$`hsa00230 Purine metabolism`
```

```
[1] "100" "10201" "10606" "10621" "10622" "10623" "107" "10714"
[9] "108" "10846" "109" "111" "11128" "11164" "112" "113"
[17] "114" "115" "122481" "122622" "124583" "132" "158" "159"
[25] "1633" "171568" "1716" "196883" "203" "204" "205" "221823"
[33] "2272" "22978" "23649" "246721" "25885" "2618" "26289" "270"
[41] "271" "27115" "272" "2766" "2977" "2982" "2983" "2984"
[49] "2986" "2987" "29922" "3000" "30833" "30834" "318" "3251"
[57] "353" "3614" "3615" "3704" "377841" "471" "4830" "4831"
[65] "4832" "4833" "4860" "4881" "4882" "4907" "50484" "50940"
[73] "51082" "51251" "51292" "5136" "5137" "5138" "5139" "5140"
[81] "5141" "5142" "5143" "5144" "5145" "5146" "5147" "5148"
[89] "5149" "5150" "5151" "5152" "5153" "5158" "5167" "5169"
[97] "51728" "5198" "5236" "5313" "5315" "53343" "54107" "5422"
[105] "5424" "5425" "5426" "5427" "5430" "5431" "5432" "5433"
[113] "5434" "5435" "5436" "5437" "5438" "5439" "5440" "5441"
[121] "5471" "548644" "55276" "5557" "5558" "55703" "55811" "55821"
[129] "5631" "5634" "56655" "56953" "56985" "57804" "58497" "6240"
[137] "6241" "64425" "646625" "654364" "661" "7498" "8382" "84172"
[145] "84265" "84284" "84618" "8622" "8654" "87178" "8833" "9060"
[153] "9061" "93034" "953" "9533" "954" "955" "956" "957"
[161] "9583" "9615"
```

The main `gage()` function requires a named vector of fold changes, where the names of the values are the Entrez gene IDs.

Note that we used the `mapIDs()` function above to obtain Entrez gene IDs (stored in `resentrez`) and we have the fold change results from *DESeq2* analysis (stored in `reslog2FoldChange`).

```
foldchanges = res$log2FoldChange
names(foldchanges) = res$entrez
head(foldchanges)
```

```
      1266      54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288
```

Now, let's run the **gage** pathway analysis.

```
keggres = gage(foldchanges, gsets=kegg.sets.hs)
attributes(keggres)
```

```
$names
[1] "greater" "less"    "stats"
```

```
head(keggres$less)
```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	7.077982e-06	-4.432593	7.077982e-06
hsa03030 DNA replication	9.424076e-05	-3.951803	9.424076e-05
hsa03013 RNA transport	1.160132e-03	-3.080629	1.160132e-03
hsa04114 Oocyte meiosis	2.563806e-03	-2.827297	2.563806e-03
hsa03440 Homologous recombination	3.066756e-03	-2.852899	3.066756e-03
hsa00010 Glycolysis / Gluconeogenesis	4.360092e-03	-2.663825	4.360092e-03

	q.val	set.size	exp1
hsa04110 Cell cycle	0.001160789	124	7.077982e-06
hsa03030 DNA replication	0.007727742	36	9.424076e-05
hsa03013 RNA transport	0.063420544	149	1.160132e-03
hsa04114 Oocyte meiosis	0.100589607	112	2.563806e-03
hsa03440 Homologous recombination	0.100589607	28	3.066756e-03
hsa00010 Glycolysis / Gluconeogenesis	0.119175854	65	4.360092e-03

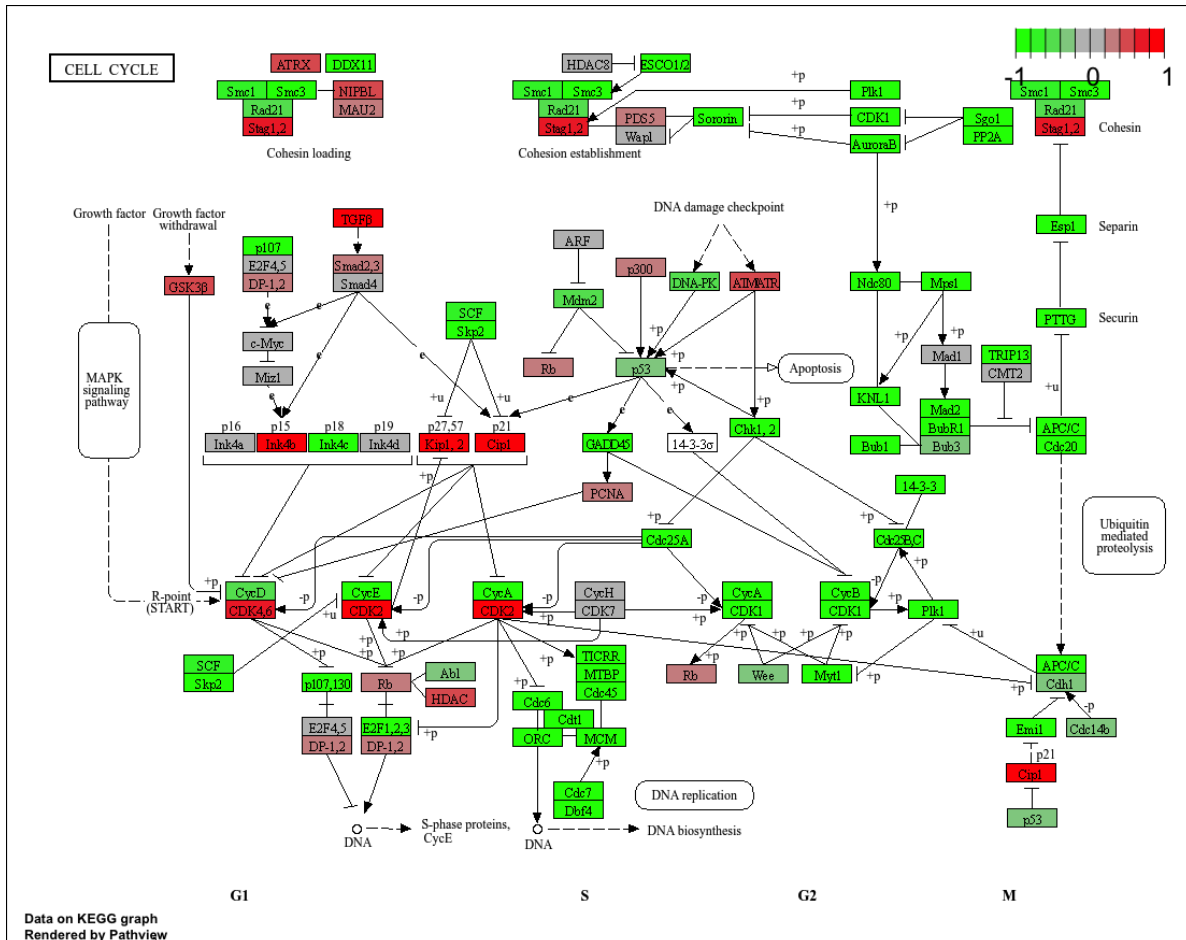
Now, let's try out the **pathview()** function from the **pathview** package to make a pathway plot with our RNA-Seq expression results shown in color. To begin with let's manually supply a **pathway.id** (namely the first part of the "hsa04110 Cell cycle") that we could see from the print out above.

```
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/aaravprasad/Desktop/BIMM143/class14

Info: Writing image file hsa04110.pathview.png



Now, let's process our results a bit more to automatically pull out the top 5 upregulated pathways, then further process that just to get the pathway IDs needed by the `pathview()` function. We'll use these KEGG pathway IDs for pathview plotting below.

```
## Focus on top 5 upregulated pathways here for demo purposes only
keggrespathways <- rownames(keggres$greater)[1:5]

# Extract the 8 character long IDs part of each string
keggresids = substr(keggrespathways, start=1, stop=8)
keggresids
```

```
[1] "hsa04740" "hsa04640" "hsa00140" "hsa04630" "hsa04976"
```

```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/aaravprasad/Desktop/BIMM143/class14
```

```
Info: Writing image file hsa04740.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/aaravprasad/Desktop/BIMM143/class14
```

```
Info: Writing image file hsa04640.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/aaravprasad/Desktop/BIMM143/class14
```

```
Info: Writing image file hsa00140.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

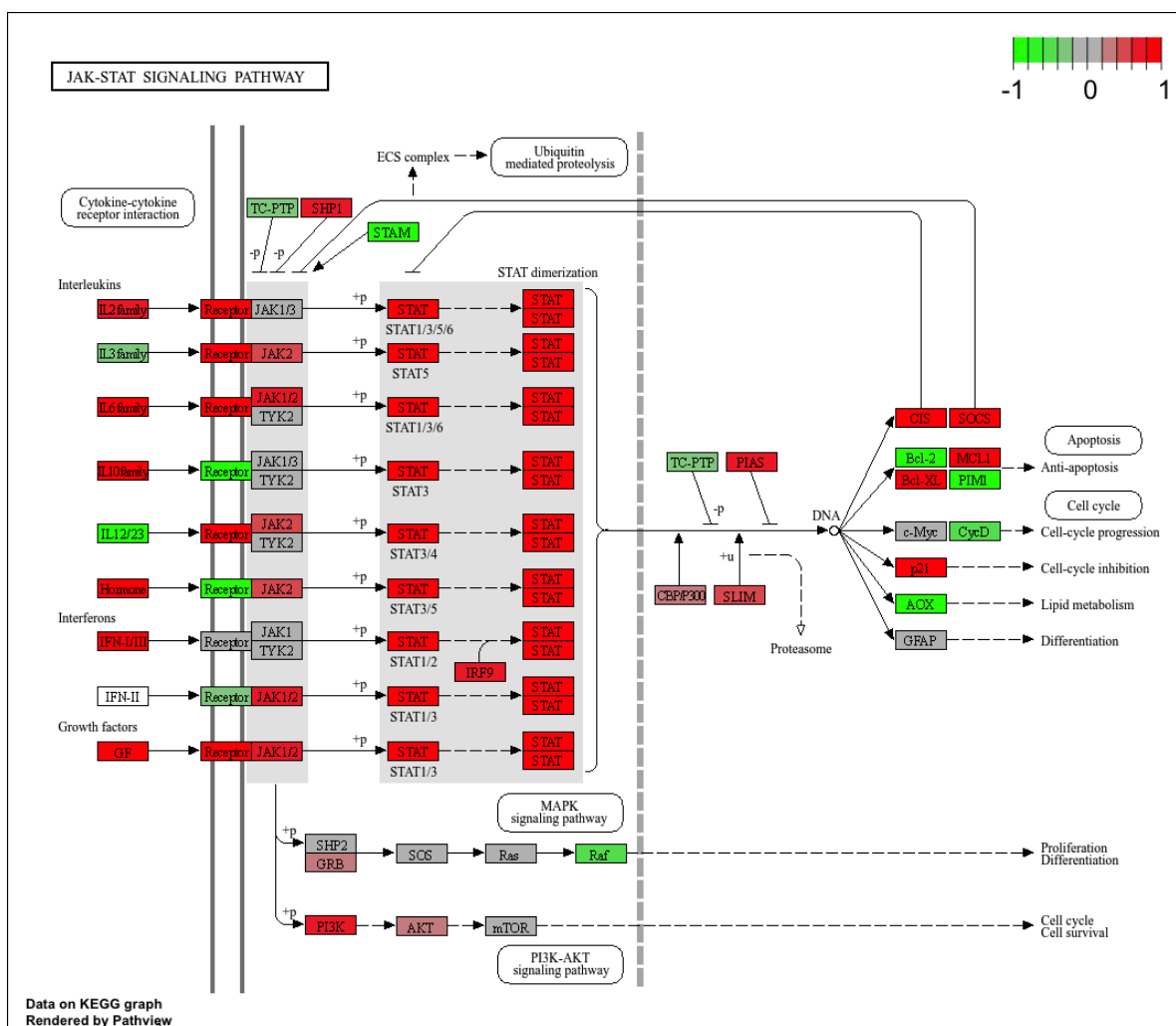
```
Info: Working in directory /Users/aaravprasad/Desktop/BIMM143/class14
```

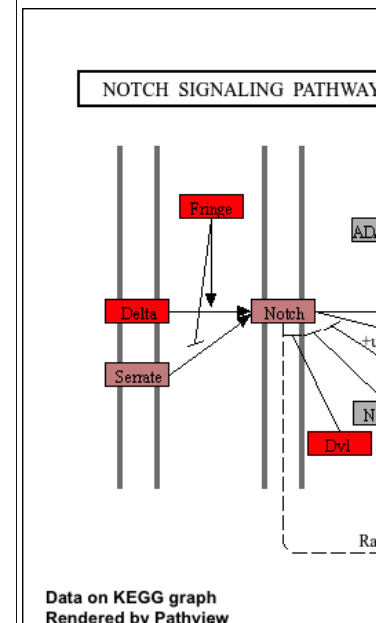
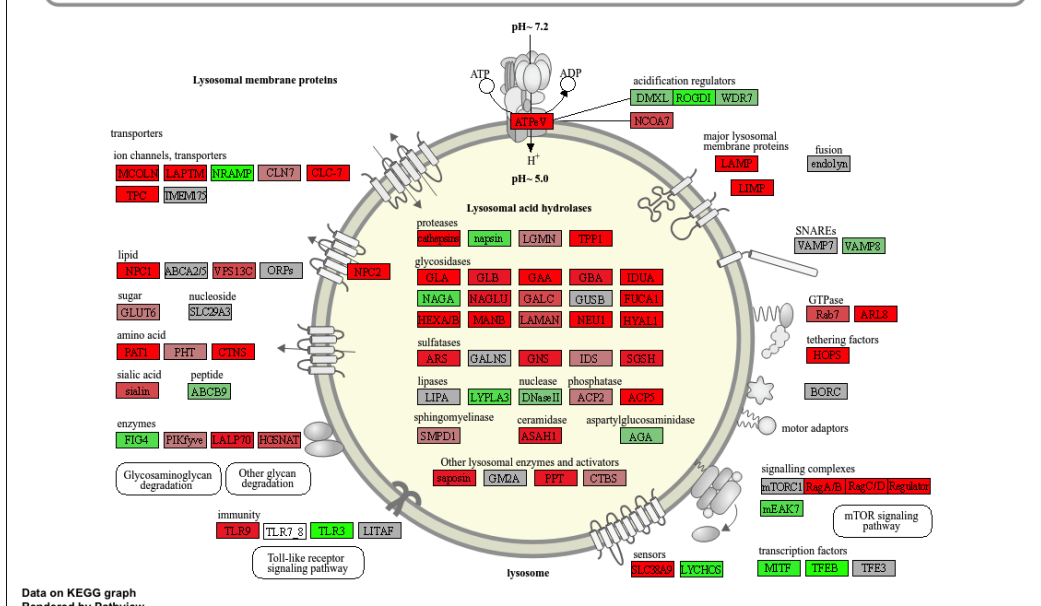
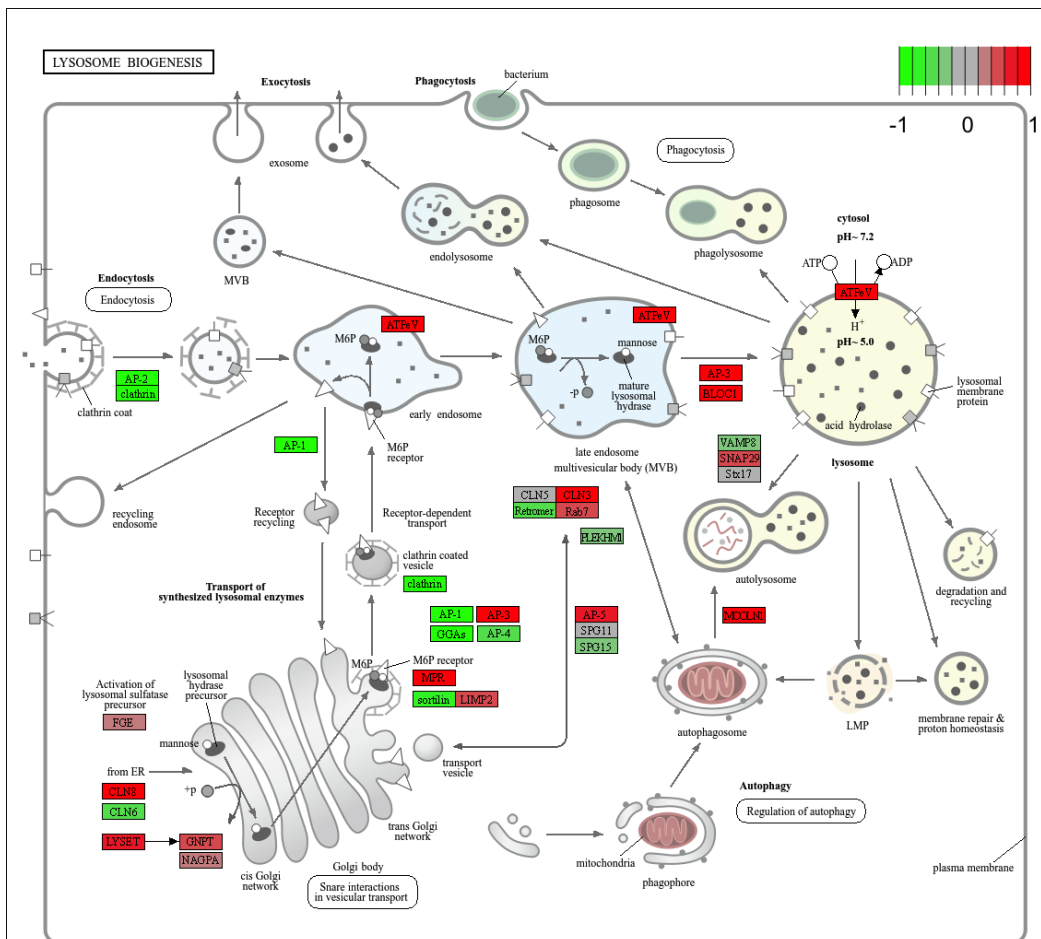
```
Info: Writing image file hsa04630.pathview.png
```

```
'select()' returned 1:1 mapping between keys and columns
```

```
Info: Working in directory /Users/aaravprasad/Desktop/BIMM143/class14
```

```
Info: Writing image file hsa04976.pathview.png
```



>Q. Can you do the same procedure as above to plot the pathview figures for the top 5 down-regulated pathways?

```
keggrespathways <- rownames(keggres$less)[1:5]

keggresids <- substr(keggrespathways, start = 1, stop = 8)
keggresids
```

```
[1] "hsa04110" "hsa03030" "hsa03013" "hsa04114" "hsa03440"
```

```
pathview(gene.data = foldchanges, pathway.id = keggresids, species = "hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/aaravprasad/Desktop/BIMM143/class14

Info: Writing image file hsa04110.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/aaravprasad/Desktop/BIMM143/class14

Info: Writing image file hsa03030.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/aaravprasad/Desktop/BIMM143/class14

Info: Writing image file hsa03013.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/aaravprasad/Desktop/BIMM143/class14

Info: Writing image file hsa04114.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/aaravprasad/Desktop/BIMM143/class14

Info: Writing image file hsa03440.pathview.png

Gene Ontology (GO)

We can also do a similar procedure with gene ontology. Similar to above, `go.sets.hs` has all GO terms. `go.subs.hs` is a named list containing indexes for the BP, CC, and MF ontologies. Let's focus on BP (a.k.a Biological Process) here.

```
data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)

lapply(gobpres, head)
```

\$greater

	p.geomean	stat.mean	p.val
GO:0007156 homophilic cell adhesion	1.734864e-05	4.210777	1.734864e-05
GO:0048729 tissue morphogenesis	5.407952e-05	3.888470	5.407952e-05
GO:0002009 morphogenesis of an epithelium	5.727599e-05	3.878706	5.727599e-05
GO:0030855 epithelial cell differentiation	2.053700e-04	3.554776	2.053700e-04
GO:0060562 epithelial tube morphogenesis	2.927804e-04	3.458463	2.927804e-04
GO:0048598 embryonic morphogenesis	2.959270e-04	3.446527	2.959270e-04

	q.val	set.size	expl
GO:0007156 homophilic cell adhesion	0.07584825	137	1.734864e-05
GO:0048729 tissue morphogenesis	0.08347021	483	5.407952e-05
GO:0002009 morphogenesis of an epithelium	0.08347021	382	5.727599e-05
GO:0030855 epithelial cell differentiation	0.16449701	299	2.053700e-04
GO:0060562 epithelial tube morphogenesis	0.16449701	289	2.927804e-04
GO:0048598 embryonic morphogenesis	0.16449701	498	2.959270e-04

\$less

	p.geomean	stat.mean	p.val
GO:0048285 organelle fission	6.626774e-16	-8.170439	6.626774e-16
GO:0000280 nuclear division	1.797050e-15	-8.051200	1.797050e-15

G0:0007067	mitosis	1.797050e-15	-8.051200	1.797050e-15
G0:0000087	M phase of mitotic cell cycle	4.757263e-15	-7.915080	4.757263e-15
G0:0007059	chromosome segregation	1.081862e-11	-6.974546	1.081862e-11
G0:0051301	cell division	8.718528e-11	-6.455491	8.718528e-11
		q.val	set.size	exp1
G0:0048285	organelle fission	2.618901e-12	386	6.626774e-16
G0:0000280	nuclear division	2.618901e-12	362	1.797050e-15
G0:0007067	mitosis	2.618901e-12	362	1.797050e-15
G0:0000087	M phase of mitotic cell cycle	5.199689e-12	373	4.757263e-15
G0:0007059	chromosome segregation	9.459800e-09	146	1.081862e-11
G0:0051301	cell division	6.352901e-08	479	8.718528e-11

\$stats

	stat.mean	exp1
G0:0007156 homophilic cell adhesion	4.210777	4.210777
G0:0048729 tissue morphogenesis	3.888470	3.888470
G0:0002009 morphogenesis of an epithelium	3.878706	3.878706
G0:0030855 epithelial cell differentiation	3.554776	3.554776
G0:0060562 epithelial tube morphogenesis	3.458463	3.458463
G0:0048598 embryonic morphogenesis	3.446527	3.446527

Reactome Analysis

Reactome is database consisting of biological molecules and their relation to pathways and processes.

Let's now conduct over-representation enrichment analysis and pathway-topology analysis with Reactome using the previous list of significant genes generated from our differential expression results above.

First, using R, output the list of significant genes at the 0.05 level as a plain text file:

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8146"
```

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Q: What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

Cell Cycle / Cell Cycle, Mitotic has the most significant Entities p-value. This matches the KEGG results, where Cell cycle was the top down-regulated pathway. Differences can come from Reactome vs. KEGG using different pathway definitions, gene IDs, pathway sizes, and statistical methods.

Save our results